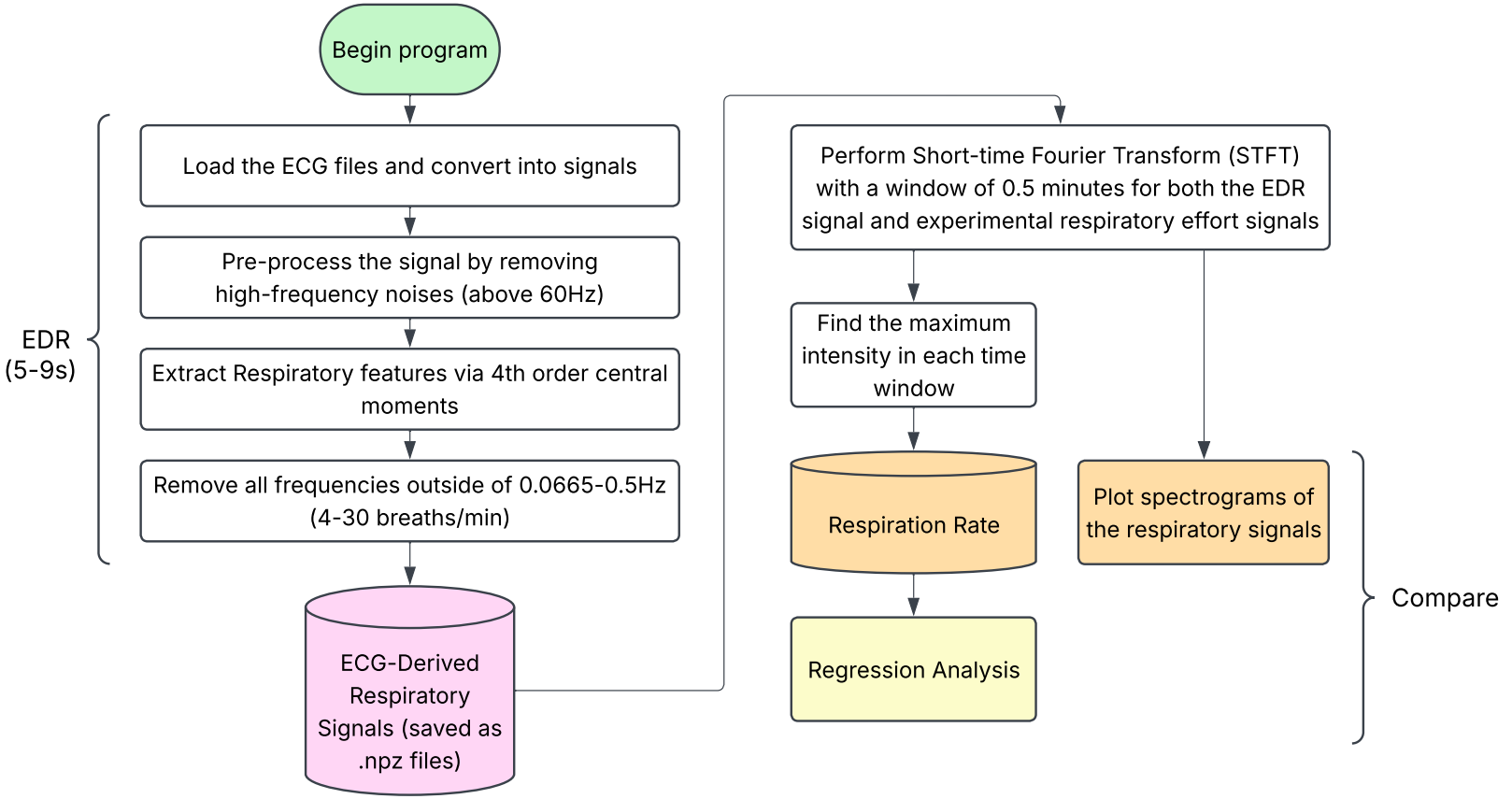
Begin program
Load the ECG files and convert into signals
Pre-process the signal by removing high-frequency noises (above 60Hz)
Extract Respiratory features via 4th order central moments
Remove all frequencies outside of 0.0665-0.5Hz (4-30 breaths/min)
EDR (5-9s)
ECG-Derived Respiratory Signals (saved as .npz files)
Perform Short-time Fourier Transform (STFT) with a window of 0.5 minutes for both the EDR signal and experimental respiratory effort signals
Find the maximum intensity in each time window
Respiration Rate
Regression Analysis
Plot spectrograms of the respiratory signals
Compare